Course Project Proposal on

Real-Time Dynamic Path Planning via Learned First-Order Perturbations of Laplacian Potential Fields

Boxiang (William) Fu

16-832 (Spring 2026) Integrated Planning and Learning
Carnegie Mellon University

02/18/2026

Motivation

Two equally valid descriptions of the universe...

Newtonian Mechanics

- Local approach
- Mediated by Force particles
- Solves for cause-and-effect one step at a time

Lagrangian Mechanics

- Global approach
- Mediated by Force fields
- Solves for entire trajectory using the Principle of Least Action

Robotics Application

- Dijkstra's and A* algorithms
- Graph traversal algorithms

Robotics Application

- Navigation functions
- Laplacian potential fields

Problem Statement

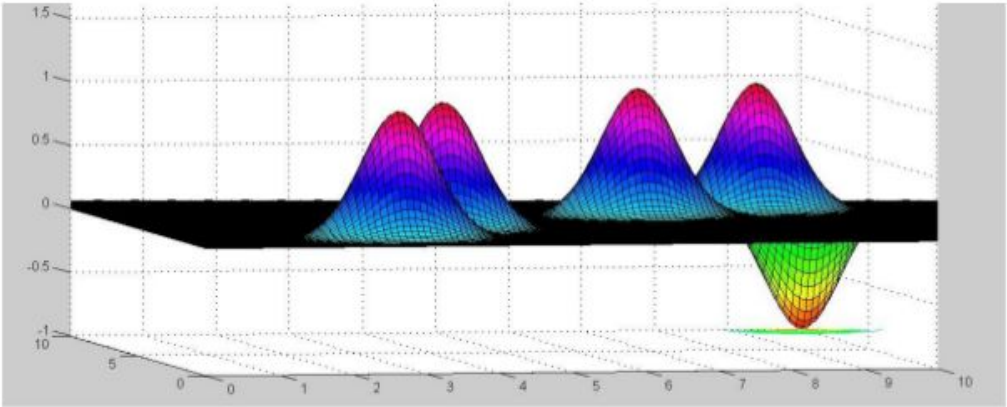
- Given an initial static map with unknown dynamic obstacles, solve the entire (global) trajectory in real-time to the goal location
- Use case: Shopping malls with known floor layout and dynamic pedestrians. Robot must reach goal location while avoiding pedestrians

Related Work

- Path planning via navigation functions (Rimon & Koditschek, 1988)
- Laplace's equation as navigation function (Connolly et al., 1990)
- Heat transfer equation as navigation function (Wang and Chirikjian, 2000)
- Path planning via Hamiltonians (Casagrande et al., 2017)
- Artificial potential fields (Wu et al., 2023)
- Dynamic path planning via perturbed Laplace's equation (Ours)

Proposed Methodology (Static Map)

- Model static obstacles and boundaries as “positively charged” source particles
- Model goal location as “negatively charged” sink particle
- These particles form Dirichlet boundary conditions on free-space
- Solve for Laplace’s equation in free-space ($\nabla^2\varphi(x) = 0$)
- Robot will follow the path of least resistance ($dx/dt = -\nabla\varphi(x)$)



Proposed Methodology (Dynamic Map)

- Look for disturbances in the local neighborhood of the robot
- Calculate the change in the map
- Model additional dynamic obstacles as charged particles
- Use neural network to estimate the first-order perturbation in the potential field due to the new particles
- Superimpose the first-order perturbation with the original potential field
- Robot will follow the (new) path of least resistance ($dx/dt = -\nabla\varphi(x)$)
- Repeat for next timestep

Useful Properties

- Given some reasonable assumptions (e.g. Cauchy complete, continuous space, etc…), the potential field from Laplace's equation has some properties:
- **Max/Min Principle**: Maxima or minima can only occur at the boundary
  - Robot cannot get stuck at a local minimum in the map
- **Uniqueness**: Solution field is unique
  - Finding one potential map is sufficient
- **Analytic**: First-order perturbation is guaranteed to exist
  - Permits Taylor/Laurent expansion in neighborhood of robot
- **Superposition**: Perturbed field can be added linearly to original field
  - Saves computation by not having to recalculate entire field

Initial Thoughts

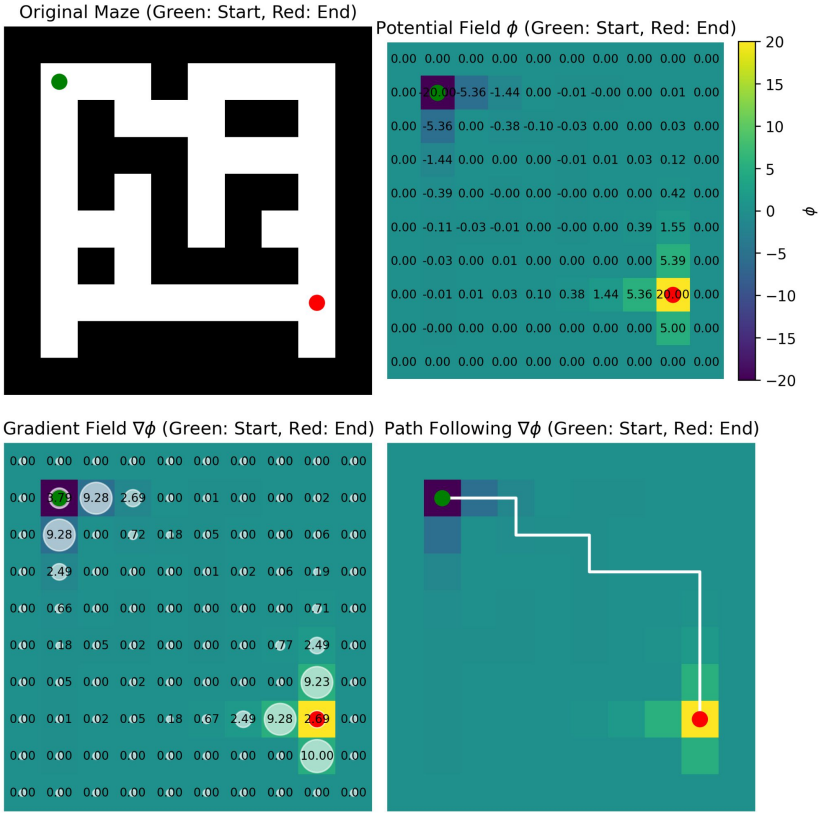
- Idea theoretically sound in continuous Euclidean space, but could run into problems using a discretized grid map
- Local minimums could exist once we discretize the map (robot may get stuck)
- Data for neural network estimate may need to be hand-crafted
- First order perturbation may not be enough to approximate new potential field
- Project scope might be too large…

Deliverables

- (Week 7) A path planning algorithm using a Laplacian potential field in a static 2D environment (reproduce Connolly et al. [1990])
- (Week 8) A path following algorithm that follows the path of least resistance
- (Week 9) An object detection algorithm that detects obstacles around the robot
- (Week 11) A learned neural network that estimates the first-order perturbation in the potential field due to nearby obstacles
- (Week 13) A dynamic path planning algorithm using the perturbed Laplacian potential field in a dynamic 2D environment
- (Stretch Goal) Implement the algorithm on a real robot in a dynamic environment
- (Stretch Goal) Extend the algorithm to 3D and/or abstract non-Euclidean vector spaces (e.g. configuration space)

Test Domain (Preliminary)

- Preliminary implementation on 2D static map
- Robot modelled as omnidirectional in 2D grid world
- Custom 2D grid world simulator
- Potential field calculated using the relaxation method
- Path planning algorithm correctly gives shortest path from start to end
- Code adapted from Swayam Singh (https://github.com/SwayamInSync/Laplace-Maze-Solver)



Thank You!